

DOM de H_3 trigonal

❖ Ce qui a été vu peut-être simplement étendu à n'importe quel fragment H_n , pour H_3 :

$$\begin{pmatrix} \alpha - \varepsilon & \beta & \beta \\ \beta & \alpha - \varepsilon & \beta \\ \beta & \beta & \alpha - \varepsilon \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = 0 = x \begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 1 & x \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ x & 1 \end{vmatrix} = x(x^2 - 1) - 2x + 2$$

$$= x^3 - x - 2x + 2 = x^3 - 3x + 2$$

$$= (x - 1)^2(x + 2)$$

$$\begin{cases} c_1 + c_2 + c_3 = 0 \\ c_1 + c_2 + c_3 = 0 \\ c_1 + c_2 + c_3 = 0 \end{cases} \implies c_1 + c_2 + c_3 = 0$$

$$\phi_3 = c_1\chi_1 + c_2\chi_2 + c_3\chi_3$$

$$\langle \phi_3 | \phi_2 \rangle = 0 \iff c_2 = c_3$$

$$\phi_3 = c_1\chi_1 + c_2(\chi_2 + \chi_3)$$

$$\langle \phi_3 | \phi_1 \rangle = 0 \iff c_1 = -2c_2$$

$$\phi_3 = -2c_2\chi_1 + c_2(\chi_2 + \chi_3)$$

$$\phi_3 = \pm \frac{1}{\sqrt{6}} (-2\chi_1 + \chi_2 + \chi_3)$$

